Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka

AI Conversational Travel Analyzer with Intelligent Package Generation and Real-Time Adaptive Engine

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Project Proposal Report

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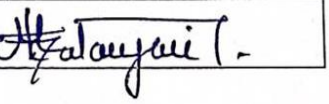
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Tourism planning in Sri Lanka can be difficult because travelers need to arrange flights, hotels, transport, and activities separately. Most existing travel systems react only when a user asks a question. However, most existing travel planning systems lack the ability to understand complex multi-intent travel requests and generate optimized travel packages automatically. In addition, current systems rarely integrate real-time environmental factors such as weather, traffic, and travel disruptions into itinerary planning. This creates a gap in intelligent tourism systems that can provide proactive, adaptive, and personalized travel planning support. They do not automatically understand complex requests, manage budgets, or check for real-time disruptions.

This research proposes an AI Conversational Travel Analyzer, Intelligent Package Generator, and Real-Time Adaptive Engine (Tri-Agent System) that creates and manages a completely personalized vacation package.

First, the Multi-Intent NLP Analyzer uses Large Language Models (LLMs) like Llama 3 or GPT-4o to understand natural language input. The Multi-Intent NLP Analyzer uses Large Language Models (LLMs) such as Llama-3 or GPT-4o to interpret natural language travel requests and extract key travel constraints including destinations, budget limits, travel duration, and transportation preferences.

Second, the Intelligent Package Generator applies NSGA-II (Non-dominated Sorting Genetic Algorithm II), a multi-objective evolutionary optimization algorithm to solve a Traveler Constraint Satisfaction Problem. It selects the best combination of flights, hotels, and transport options within the 4 lakh budget. It also takes into account Sri Lanka's varying travel conditions, where travel time changes based on road type and terrain. The system improves both cost and time while maximizing comfort and practicality.

Third, the Real-Time Adaptive Engine keeps track of live APIs like weather updates, flight status, traffic conditions, and crowd levels. If a disruption happens, such as heavy rainfall in Galle or a train delay, the system automatically adjusts the itinerary. A Proactive Voice Agent then informs the traveler and suggests alternative destinations or updated travel plans before the disruption impacts the trip.

The main innovation of this research is moving from reactive chatbot systems to proactive travel intelligence. Unlike traditional systems that only respond after problems arise, this system actively monitors risks and automatically adjusts the itinerary. It also accounts for travel constraints specific to Sri Lanka, which are often overlooked by global travel planners.

This research shows how intelligent AI can improve safety, convenience, and decision-making in tourism systems. The proposed system can improve tourism planning efficiency and enhance traveler safety by providing intelligent and adaptive travel recommendations. From a commercial perspective, such technology can support tourism businesses, travel agencies, and digital travel platforms by enabling automated travel package generation and real-time travel assistance. This contributes to the development of smarter tourism services in Sri Lanka.

Keywords: Agentic AI, Multi-Intent NLP, Genetic Algorithm, Travel Optimization, Real-Time Monitoring

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
DSR	Design Science Research
GA	Genetic Algorithm
LLM	Large Language Model
NLP	Natural Language Processing
SDG	Sustainable Development Goal
UI	User Interface
UX	User Experience
WBS	Work Breakdown Structure

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1. INTRODUCTION

1.1 Background and Context

Tourism planning is an important activity for international travelers visiting destinations such as Sri Lanka. A traveler must organize several travel components including flights, accommodation, transportation, and tourist activities. Managing these arrangements separately across multiple online platforms can be time-consuming and difficult. For example, a traveler visiting Sri Lanka from Dubai with a limited budget must search for flights, compare hotel prices, arrange transport between cities, and schedule tourist activities while ensuring the travel plan fits within the available budget and travel duration. This process often requires using multiple websites and applications, which increases the complexity of travel planning.

This issue affects both **international tourists and local tourism service providers**. Travelers often face difficulties in coordinating travel arrangements efficiently, especially when visiting multiple destinations within a limited time. At the same time, tourism businesses such as hotels, transport providers, and travel agencies rely on effective digital systems to promote their services and attract travelers. Inefficient travel planning tools can reduce the overall tourism experience and limit the ability of travelers to explore multiple destinations within a country.

The importance of improving travel planning systems is evident from global tourism statistics. According to the **United Nations World Tourism Organization (UNWTO)**, international tourist arrivals exceeded **900 million worldwide**, demonstrating the increasing demand for digital tourism services. In Sri Lanka, tourism is a key contributor to the national economy. According to the **Sri Lanka Tourism Development Authority (SLTDA)**, the country welcomed **over 1.5 million international tourists in recent years**, generating significant revenue and supporting employment within the tourism sector. As tourism continues to grow, intelligent digital systems are required to support travelers in planning efficient and personalized trips.

Several existing digital travel platforms already support travel planning activities. Popular applications such as Google Maps, TripAdvisor, and Expedia provide services such as route planning, hotel recommendations, and travel reviews. Some modern platforms also provide chatbot assistants that allow users to ask questions and receive travel suggestions. While these platforms offer useful information, they still have several limitations. Most systems require users to manually search for travel options and combine multiple services such as flights, accommodation, and transport. In addition, many existing travel assistants provide only text-based recommendations and do not perform optimization to balance travel constraints such as budget, travel time, and transportation availability.

Travel planning is particularly challenging in Sri Lanka due to several location-specific factors. The country's transportation network includes mountainous roads, varying travel speeds, and different terrain conditions that can significantly affect travel time. In addition, weather changes, train delays, and seasonal tourist crowd levels can disrupt travel plans. Current global travel planning systems rarely consider these regional conditions when generating travel itineraries. As a result, travelers often need to manually adjust their plans when unexpected disruptions occur.

To address these limitations, this research proposes an **AI-based proactive travel planning system** designed specifically for Sri Lanka-based travel destinations. The proposed system consists of three main components: a **Multi-Intent NLP Analyzer**, an **Intelligent Package Generator**, and a **Real-Time**

Adaptive Engine. The Multi-Intent NLP Analyzer uses large language models to understand complex natural language travel requests and extract structured travel constraints such as budget, preferred destinations, travel duration, and transportation preferences. The Intelligent Package Generator uses optimization techniques such as Genetic Algorithms to generate a complete travel package that balances cost, travel time, and user preferences. The Real-Time Adaptive Engine continuously monitors environmental and transportation conditions, such as weather updates and traffic disruptions, and automatically updates the travel itinerary when necessary.

This research also supports **United Nations Sustainable Development Goal (SDG) 8 – Decent Work and Economic Growth**, which promotes sustainable tourism development through technological innovation. By improving travel planning efficiency and supporting personalized tourism experiences, the proposed system can contribute to the growth of the tourism sector and enhance the overall travel experience in Sri Lanka.

The proposed study falls under the **Artificial Intelligence and Intelligent Systems research cluster**, as it focuses on the development of an intelligent travel planning system using Natural Language Processing, optimization algorithms, and real-time data processing techniques to improve tourism decision-making and itinerary generation.

1.2 Literature Review

Artificial Intelligence (AI) technologies are increasingly used in the tourism industry to enhance travel planning, recommendation systems, and customer support services. With the rapid growth of digital tourism platforms, intelligent systems are being developed to assist travelers in discovering destinations, planning itineraries, and managing travel information efficiently.

According to Buhalis and Law, information technology has significantly transformed tourism management by enabling digital booking systems, online travel agencies, and personalized travel services [1]. Modern tourism platforms allow travelers to search for destinations, compare hotel prices, and plan trips through digital platforms. These technological advancements have improved accessibility and efficiency in travel planning.

The concept of **smart tourism** has also gained attention in recent years. Gretzel et al. describe smart tourism systems as technology-driven platforms that integrate artificial intelligence, mobile computing, and big data analytics to enhance tourist experiences [1] . These systems collect and analyze traveler data such as preferences, location information, and travel history to provide personalized recommendations and support travel decision-making.

Artificial Intelligence techniques such as **machine learning, natural language processing (NLP), and knowledge-based systems** are widely used in modern tourism applications. Russell and Norvig explain that AI systems can analyze large amounts of data, recognize patterns, and provide intelligent recommendations to assist users in decision-making processes [2]. These technologies are commonly used in chatbot assistants and recommendation engines in travel platforms.
Recent studies highlight the growing use of AI in travel recommendation systems. Wang et al. demonstrate that AI-based tourism platforms can analyze traveler preferences, travel history, and user-generated content such as online reviews to recommend destinations and tourist attractions [3]. These systems help travelers identify suitable travel locations and activities based on their interests.
In addition to academic research, several commercial travel platforms already use AI technologies.

Applications such as **Google Maps, TripAdvisor, and Expedia** provide services including route planning, hotel recommendations, and travel reviews. Some modern platforms also integrate conversational assistants that allow users to search for travel information using natural language queries. These systems improve accessibility and user experience in travel planning.

Another important development in tourism technology is the use of **big data and text analytics**. Xiang and Sigala explain that tourism platforms collect large amounts of traveler-generated data such as reviews, ratings, and social media posts, which can be analyzed to understand tourist behavior and improve recommendation systems [4]. These data-driven approaches allow tourism platforms to provide more personalized travel suggestions.

Despite these technological advancements, several limitations still exist in current tourism planning systems. Most travel platforms operate in a **reactive manner**, meaning that the system provides suggestions only after the user manually searches for travel options. These systems rarely monitor travel conditions continuously or update travel plans automatically when disruptions occur.
In addition, many generative AI travel assistants can generate travel itineraries in text format but lack mathematical optimization techniques to balance important travel constraints such as **budget limitations, travel time, transportation availability, and user comfort**. As a result, travelers must manually compare different travel options before selecting a final itinerary.

Another limitation is that global travel planning systems are not specifically designed to consider **region-specific travel conditions**, such as mountainous terrain, varying road speeds, and seasonal travel disruptions commonly experienced in destinations like Sri Lanka. These factors can significantly affect travel time and itinerary feasibility, yet they are often not incorporated into travel planning algorithms. Furthermore, existing tourism systems rarely integrate multiple intelligent capabilities such as **conversational natural language understanding, travel package optimization, and real-time monitoring of environmental conditions** within a single platform. This limitation highlights the need for more advanced intelligent tourism systems that can combine these capabilities to provide more adaptive and proactive travel planning support.

1.3 Research Gap

Existing tourism planning systems provide useful travel information and recommendations; however, they still have several limitations. Most current travel platforms operate in a reactive manner where the system responds only after the user submits a query. These systems do not proactively monitor travel conditions or automatically update travel plans when disruptions occur.

In addition, many generative AI travel assistants are capable of producing textual travel itineraries but do not apply optimization techniques to balance multiple travel constraints such as budget limitations, travel time, transportation availability, and traveler comfort. As a result, users must manually evaluate different options before selecting a final travel plan.

Another limitation is that existing global travel planning systems are not specifically designed to handle region-specific travel conditions such as those found in Sri Lanka, including mountainous terrain, varying road speeds, seasonal travel disruptions, and transportation delays.

Furthermore, most existing tourism platforms do not integrate multiple intelligent capabilities such as conversational natural language understanding, optimization algorithms for package generation, and real-time monitoring of travel disruptions within a single system. Because of these limitations, there is a need for an intelligent travel planning system that can automatically understand complex travel requests, generate optimized travel packages, and dynamically adapt travel plans based on real-time environmental and transportation conditions.

2.OBJECTIVE

2.1 Main Objective

To design and evaluate an AI-based proactive travel planning system that generates optimized personalized vacation packages and automatically adapts them based on real-time environmental and transport disruptions.

2.2 Specific Objectives

1. To develop a Multi-Intent NLP Analyzer capable of extracting structured travel constraints from free-text input.
2. To implement a Genetic Algorithm that optimizes travel packages within a fixed budget while considering time, comfort, and transport constraints.
3. To model Sri Lanka-specific travel time variations and terrain-based speed differences.
4. To design a Real-Time Adaptive Engine that monitors weather, flight status, traffic, and crowd data continuously.
5. To build a Proactive Voice Agent that notifies users and suggests alternative plans during disruptions.
6. To evaluate system performance using optimization accuracy and response-time metrics.

3. METHODOLOGY

3.1 Research Approach

This research follows the **Design Science Research (DSR) methodology**, which is widely used in information technology research when developing and evaluating innovative technological solutions.
The main objective of this research is to design and develop an **AI-based intelligent travel itinerary planning system** that improves the efficiency and personalization of travel planning.
The proposed system integrates **Natural Language Processing (NLP)** to understand user travel preferences and a **NSGA-II Multi-Objective Optimizer** to generate optimized travel itineraries based on constraints such as budget, travel duration, and destination preferences.
According to the Design Science Research methodology, the research process involves the following stages:

- **Problem Identification** – Understanding limitations in existing travel planning systems.
- **Artifact Design** – Designing the architecture of the AI-based travel planning system.
- **Artifact Development** – Implementing the system using appropriate tools and algorithms.
- **Evaluation** – Measuring system performance using quantitative metrics and comparison with existing solutions.

This methodology is appropriate because the research contribution is the **design, development, and evaluation of a novel AI-based travel planning artifact** that improves travel itinerary optimization.

3.2 Technical Methods and Algorithms

Phase 1: Multi-Intent NLP Analyzer

The system uses a Large Language Model to:

- Detect budget (400,000 LKR)
- Extract trip duration (4 days)
- Identify destinations
- Identify transport preference
- Understand travel style

This converts unstructured text into structured constraints.

Phase 2: Intelligent Package Generator

A NSGA-II Multi-Objective Optimizer performs multi-objective optimization by:

- Selecting suitable flights
- Allocating budget between flights, hotels, and transport
- Minimizing travel time
- Maximizing comfort
- Ensuring total cost ≤ 400,000 LKR

The algorithm evaluates multiple possible combinations and selects the best solution using Pareto optimization.

Phase 3: Real-Time Adaptive Engine

After the package is generated, Guardian Loop activates.
It checks:

- Weather API
- Flight status API
- Traffic conditions
- Crowd density data

If a disruption occurs (e.g., heavy rain in Galle), the system:

1. Calculates risk score
2. Re-runs optimization
3. Updates itinerary
4. Notifies user through voice

Phase 4: Proactive Voice Agent

Instead of waiting for user input, the system says:
"Due to heavy rainfall in Galle, your Day 3 plan has been updated. I recommend visiting an indoor cultural site in Colombo instead."
This demonstrates proactive AI behavior.

3.3 Tools and Platforms with Justification

The proposed system will be developed using several tools and platforms that support **mobile application development, artificial intelligence processing, and system deployment**.

Tool / Platform	**Purpose**	**Justification**
Python	Backend development and AI processing	Python provides extensive libraries for artificial intelligence, machine learning, and data processing, making it suitable for implementing NLP models and optimization algorithms.
React Native	Mobile application development	React Native allows cross-platform development for both Android and iOS using a single codebase, reducing development time and ensuring consistent

		user experience.
NSGA-II Multi-Objective Optimizer	Itinerary optimization	NSGA-II is an advanced multi-objective optimization algorithm used to generate optimal travel plans by balancing multiple factors such as cost, time, and user preferences efficiently.
Llama 3 / GPT-4o	Natural Language Processing	Large Language Models are used to analyze natural language input from users and extract structured travel preferences such as budget, destinations, and duration.
REST APIs	External data integration	APIs will be used to retrieve real-time information such as weather conditions, flight schedules, and traffic data required for travel itinerary optimization.
Amazon Web Services / Microsoft Azure	Cloud deployment	Cloud platforms provide scalable infrastructure for hosting backend services, managing APIs, and ensuring system reliability.
Firebase	Mobile backend services	Firebase can be used for user authentication, database storage, and real-time notifications within the mobile application.

Table 1 : Tools and Platforms

3.4 Data Collection and Ethical Considerations

Data required for the system will be obtained from **publicly available travel datasets and real-time travel APIs**.
These data sources may include:

- Hotel price datasets
- Flight schedules
- Tourist attraction information
- Weather conditions
- Travel distance and route data

In addition, **simulated travel planning scenarios** will be generated to evaluate the system's performance under different travel conditions such as varying budgets, trip durations, and destination combinations. Ethical considerations will also be followed throughout the research. The system will not store personal user data without consent. All testing scenarios will use **anonymized data** to ensure user privacy and compliance with data protection principles.

3.5 Validation Strategy

The performance of the proposed system will be evaluated through controlled experiments using **simulated travel planning scenarios**.

Approximately **100 travel planning scenarios** will be generated with different combinations of travel preferences such as:

- Budget levels
- Trip duration
- Destination combinations
- Transportation preferences

The system will generate travel itineraries for each scenario, and the results will be analyzed using quantitative performance metrics.

The following evaluation metrics will be used:

NLP Extraction Accuracy
Measures how accurately the system extracts travel preferences from user input.

NSGA-II Optimization Convergence Rate
Measures how quickly the NSGA-II Multi-Objective Optimizer finds optimal or near-optimal itinerary solutions while balancing multiple objectives such as cost, time, and preferences.

Budget Optimization Efficiency
Evaluates how effectively the generated itinerary satisfies the user's budget constraints.

System Response Time
Measures the time taken by the system to generate travel itinerary recommendations.

Risk Detection Accuracy
Measures the system's ability to detect disruptions such as weather changes and trigger itinerary updates.
The results will also be compared with existing travel planning platforms to demonstrate the effectiveness of the proposed system.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed system is designed as a **Tri-Agent Architecture** consisting of three main intelligent components:

1. Multi-Intent NLP Analyzer
2. Intelligent Package Generator (NSGA-II Optimization)
3. Real-Time Adaptive Engine with Proactive Voice Agent

The system works in two major phases:
Phase 1 – Personalized Package Generation
Phase 2 – Continuous Monitoring and Adaptive Re-Optimization

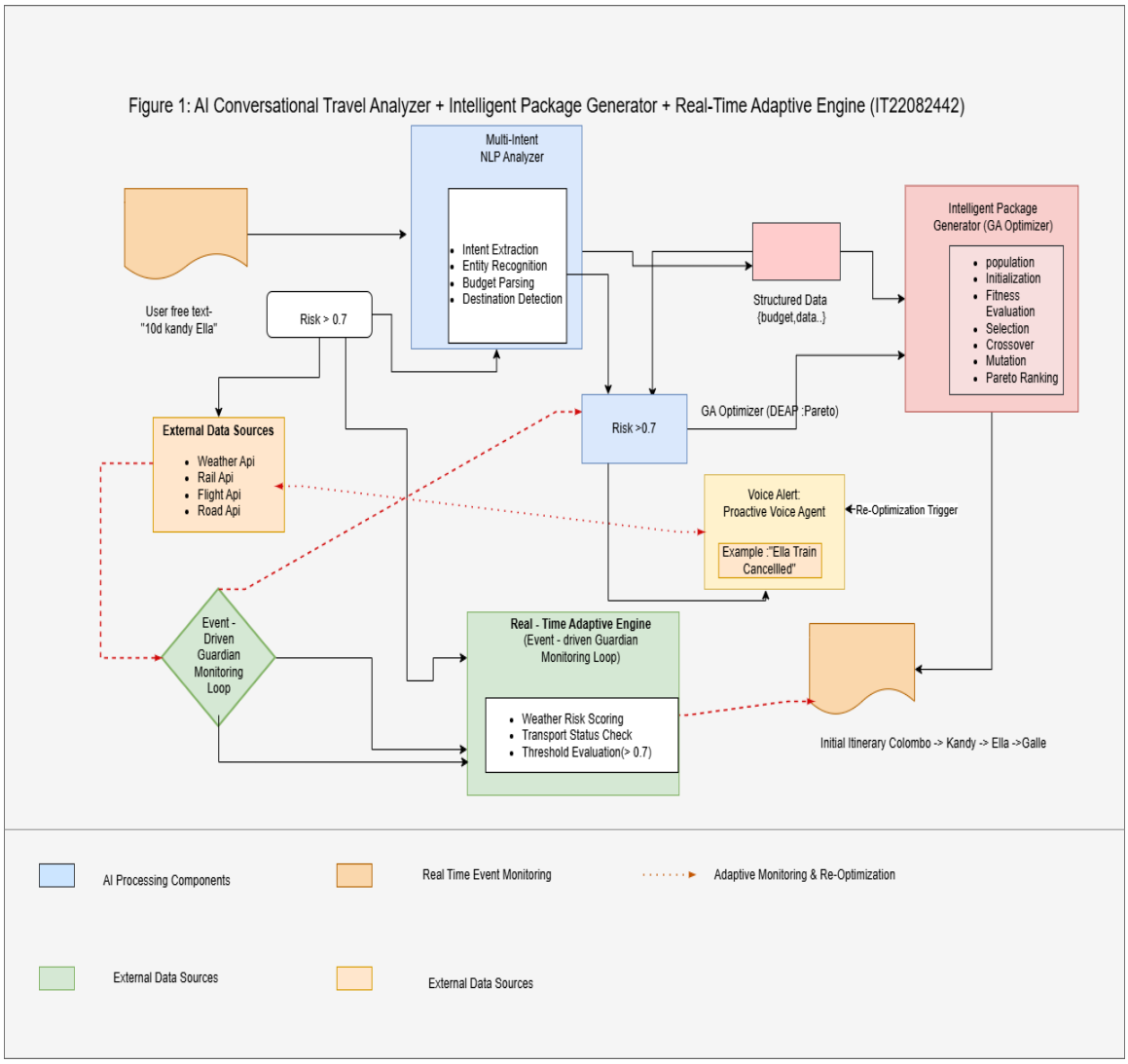


Figure 1 : AI Conversational Travel Analyzer + Intelligent Package Generator + Real - Time Adaptive Engine

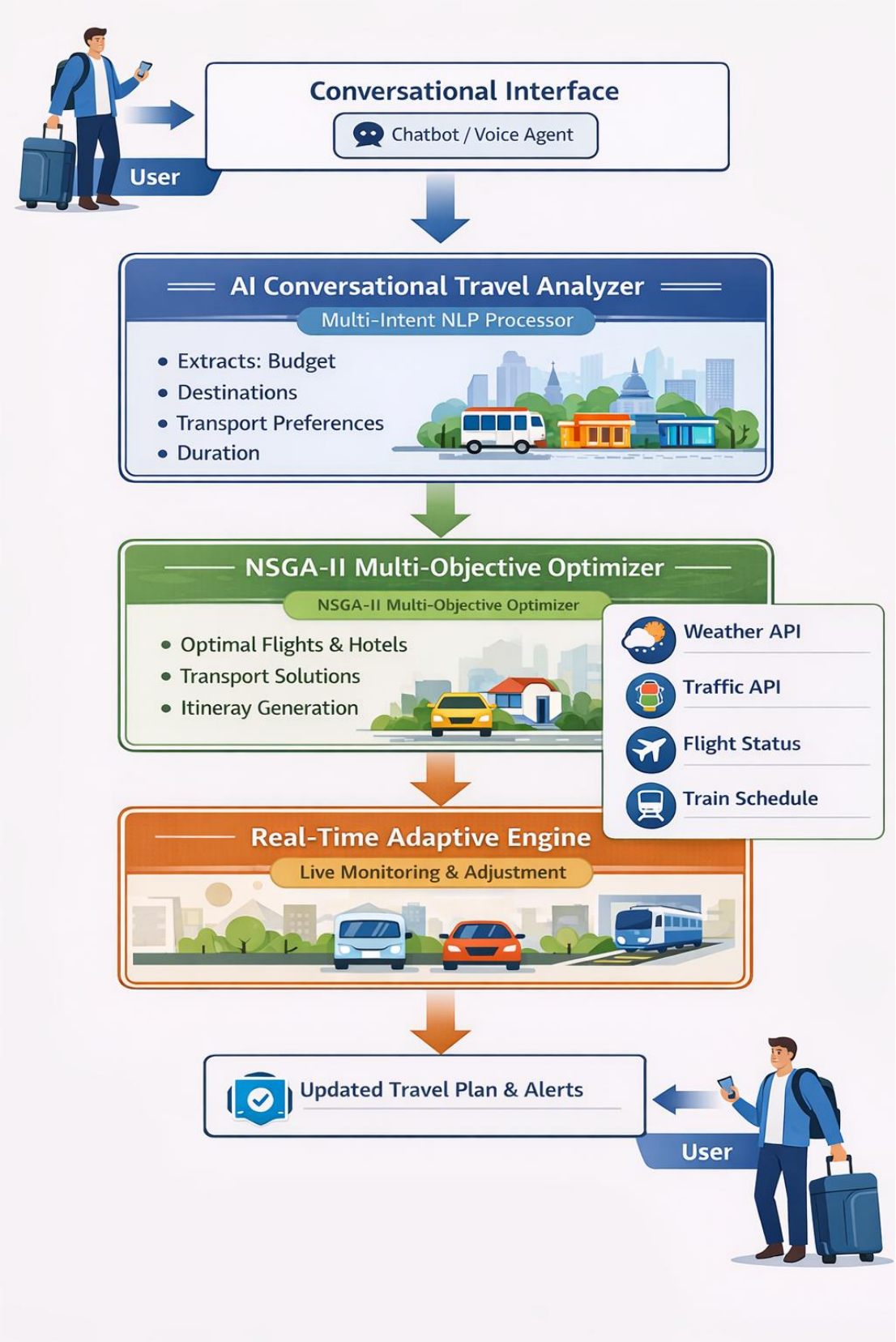


Figure 2 : AI Conversational Travel Analyzer + Intelligent Package Generator + Real - Time Adaptive Engine

4.2 System Components

Multi-Intent NLP Analyzer

This module receives natural language input from the user.
Example:
“I want a 4-day trip to Sri Lanka. Budget is 4 lakhs. I want to visit Kandy, Nuwara Eliya and Galle. I prefer traveling by car or train.”
The NLP module extracts:

- Budget
- Duration
- Destination list
- Transport preference
- Travel style

It converts unstructured text into structured constraints for optimization.

Intelligent Package Generator

This module uses a **NSGA-II Multi-Objective Optimizer** to solve the Traveler Constraint Satisfaction Problem.
It:

- Selects best flight options
- Allocates budget across flights, hotels, and transport
- Minimizes travel time
- Maximizes comfort
- Ensures total cost ≤ budget

The algorithm evaluates multiple combinations and selects the best solution using multi-objective optimization.

Real-Time Adaptive Engine

After the package is generated, this module continuously monitors:

- Weather updates
- Flight delays
- Train status
- Traffic conditions
- Crowd levels

If a risk threshold is exceeded:

- The system re-runs the optimization
- Updates itinerary
- Activates the Voice Agent

Proactive Voice Agent

The Voice Agent informs the user automatically when a change occurs.

Example:
“Due to heavy rainfall in Galle, your Day 3 plan has been updated. I recommend visiting an indoor attraction in Colombo instead.”

This makes the system proactive rather than reactive.

4.3 Data Flow

- User Input

- NLP Extraction
- Structured Constraints
- NSGA-II Multi-Objective Optimizer
- Personalized Travel Package

- Parallel Monitoring

- Risk Assessment
- Re-Optimization
- Voice Notification

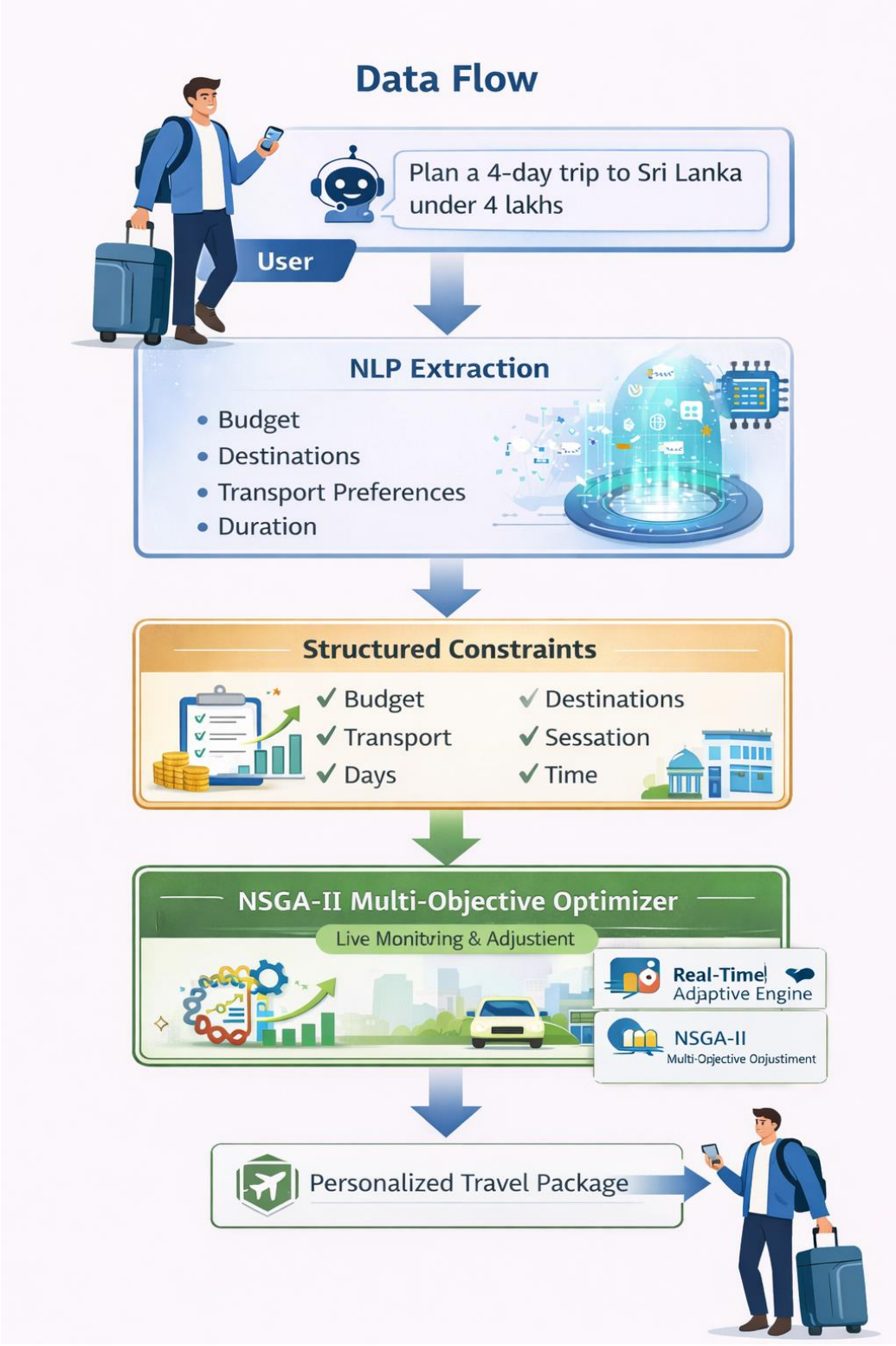


Figure 3 : Data Flow

4.4 Scalability and Security

The proposed system is designed to support scalability and secure data management to ensure reliable operation in real-world tourism environments. The architecture can be deployed using a **cloud-based microservices model**, where individual components such as the AI Conversational Travel Analyzer, Intelligent Package Generator, and Real-Time Adaptive Engine operate as independent services. This design allows the system to scale efficiently as the number of users increases, enabling travel planning services to handle large volumes of user requests without affecting system performance.

From a security perspective, several measures are considered to protect user data and system communications. All external API communications used for accessing travel information such as weather updates, traffic conditions, and transportation schedules are protected using **encrypted communication protocols such as HTTPS and secure API authentication mechanisms**. In addition, user travel preferences and personal data are stored securely using protected databases with controlled access to prevent unauthorized data usage.

Furthermore, the system design considers compliance with the **Sri Lanka Personal Data Protection Act**, ensuring that user information is handled responsibly and that privacy standards are maintained during data collection, storage, and processing. By incorporating scalable architecture and strong security mechanisms, the proposed system ensures both **efficient performance and protection of sensitive user data**.

5. USER REQUIREMENTS

5.1 Stakeholders

- International tourists
- Local tourists
- Travel agencies
- Tourism authorities

5.2 Functional Requirements

FR1 – The system shall extract budget, duration, and destination from natural language input.
FR2 – The system shall generate a complete travel package including flights, hotels, and transport.
FR3 – The system shall optimize the package within the user's budget.
FR4 – The system shall monitor live environmental and transport data.
FR5 – The system shall automatically re-optimize the itinerary when risk is detected.
FR6 – The system shall provide proactive voice alerts during disruptions.

5.3 Non-Functional Requirements

- Response time less than 5 seconds for package generation
- Secure API communication
- Scalable architecture
- High usability and simple interface
- Reliable risk detection

6. COMMERCIALIZATION PLAN

6.1 Target Market

The proposed AI-based travel planning system is primarily targeted at **international tourists visiting Sri Lanka** who require efficient tools to organize multi-destination travel itineraries. Many tourists visiting Sri Lanka explore multiple cities such as Kandy, Nuwara Eliya, and Galle, which requires careful planning of transportation, accommodation, and travel schedules. The system can assist these travelers by automatically generating optimized travel packages based on their preferences and budget.

In addition, the system can support **digital nomads**, who are professionals working remotely while traveling. These users often require flexible travel plans that can adapt to changing schedules and travel conditions. The proposed system can help them manage travel logistics more efficiently through real-time itinerary adjustments.

The platform can also benefit **travel agencies and tourism service providers** by offering an intelligent tool to generate personalized travel packages for customers. By integrating AI-based optimization and real-time monitoring capabilities, travel agencies can improve service efficiency and provide more dynamic travel recommendations to their clients.

6.2 Value Proposition

The proposed system offers a unique value proposition by functioning as a **proactive AI travel guardian** that assists travelers throughout their journey. Unlike traditional travel planning platforms that provide static recommendations, the system actively analyzes user requirements and generates **personalized travel packages** tailored to the traveler's preferences.

The platform optimizes both **travel cost and travel time** using intelligent optimization algorithms, ensuring that users receive the most efficient travel itinerary within their budget constraints. In addition, the system continuously monitors real-time travel conditions such as weather updates, transportation delays, and traffic disruptions.

When disruptions occur, the system automatically adapts the travel plan and provides updated recommendations, reducing the stress and uncertainty often associated with travel planning. This proactive approach improves the overall travel experience and allows users to focus on enjoying their trip

6.3 Revenue Model

The proposed platform can generate revenue through several sustainable business models. One primary approach is a **subscription-based service**, where users pay a monthly or annual fee to access advanced features such as AI-generated travel packages, real-time itinerary updates, and personalized travel recommendations.

Another revenue stream can be generated through **commission-based partnerships with hotels, airlines, and transportation providers**. When users book hotels, flights, or transport services through the platform, the system can receive a small commission from the service providers.

Additionally, the system can offer **API integration services for travel agencies and tourism platforms**,

allowing external companies to access the intelligent travel optimization engine. This integration can help travel agencies enhance their services by providing AI-powered travel planning tools for their customers.

6.4 Competitive Advantage

The proposed system offers several competitive advantages compared to existing travel planning platforms. One key advantage is its **Sri Lanka-specific travel modeling**, which considers regional factors such as mountainous terrain, varying road conditions, and seasonal tourism patterns. These local travel constraints are often overlooked by global travel planning systems.

Another advantage is the use of **Genetic Algorithm optimization**, which enables the system to generate highly efficient travel packages by evaluating multiple travel options and selecting the most optimal combination based on cost, travel time, and user preferences.

The system also incorporates **real-time proactive adaptation**, allowing travel itineraries to be automatically updated when disruptions such as weather changes or transportation delays occur. Furthermore, the platform provides **voice-based alerts and notifications**, enabling travelers to receive timely updates and alternative travel suggestions during their journey.

These features collectively create a more intelligent and adaptive travel planning system that enhances convenience, efficiency, and safety for travelers visiting Sri Lanka.

7.BUDGET AND JUSTIFICATION

The development of the proposed AI-based travel itinerary planning mobile application requires several resources including computing infrastructure, cloud services, development tools, and human effort. The estimated budget for the project is summarized in Table 2.

Item	Description	Estimated Cost (LKR)
Hardware Resources	Personal laptop and mobile device used for system development and testing	50,000
Software Licenses	Development tools, IDEs, and optional AI service subscriptions	15,000
Cloud Services	Cloud hosting for backend services and database deployment using platforms such as AWS or Azure	35,000
Travel Data API Access	Access to external APIs for weather data, travel information, and location services	20,000
Development and Testing Infrastructure	Mobile testing tools, debugging utilities, and system testing environments	10,000
Human Effort Cost	Time and effort spent on system design, development, testing, and documentation	60,000
Miscellaneous Expenses	Internet usage, data processing, and unforeseen development expenses	10,000
Total Estimated Cost		200,000

Table 2 : Budget and Justification

7.1Budget Justification

Hardware Resources
A laptop and mobile device are required to develop, test, and run the mobile application. These devices will support programming, AI model execution, and mobile interface testing.

Software Licenses
Although many development tools are open source, some optional AI services and development platforms may require subscriptions for extended functionality.

Cloud Services

Cloud platforms such as **Amazon Web Services** or **Microsoft Azure** will be used to deploy the backend services, manage APIs, and store travel data required for the system.

Travel Data API Access
The system relies on external data sources such as weather services and travel information APIs to generate accurate travel itineraries and provide real-time updates.

Development and Testing Infrastructure
Testing tools are required to ensure the reliability and performance of the mobile application across different devices.

Human Effort Cost
This cost represents the time and effort required for system design, development, AI model integration, testing, and documentation.

Miscellaneous Expenses
Additional costs such as internet usage, dataset processing, and other minor development-related expenses are included under this category.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown

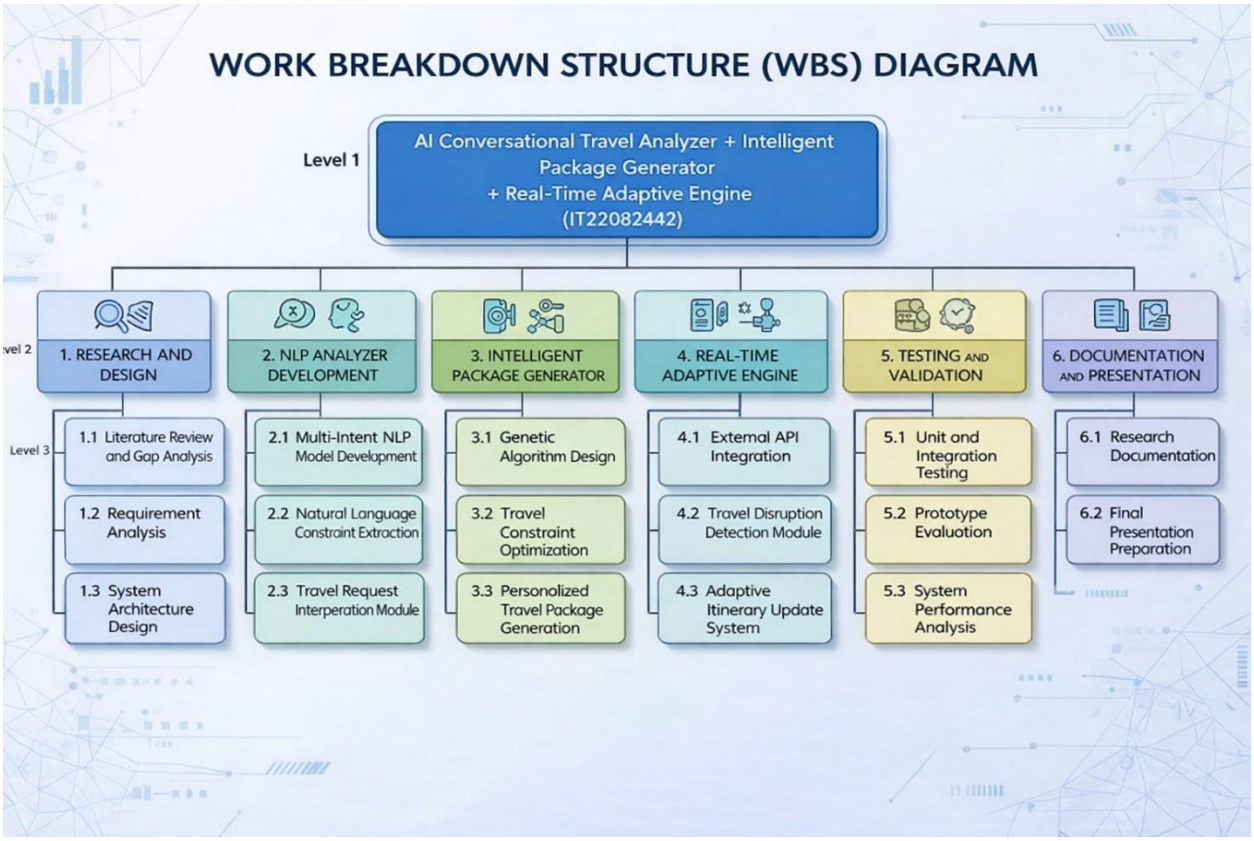


Figure 4 : Work Breakdown Structure

8.2 Timeline Alignment

The project development is scheduled according to two semester-based timelines where different components are developed in parallel before system integration.

Phase	Activity	Timeline
Phase 1	Requirement analysis, problem identification, and literature review	Semester 1 (Jan – Mar 2026)
Phase 2	Research proposal preparation and system architecture design	Proposal Presentation – March 16, 17, 18 (2026)
Phase 3	Core model development and	Semester 1 (Apr – May 2026)

	initial prototype implementation	
Phase 4	Prototype demonstration and first progress evaluation	Progress Presentation 1 – May 4, 5, 6 (2026)
Phase 5	Advanced model development, optimization, and module integration	Semester 2 (Jun – Aug 2026)
Phase 6	System testing, performance evaluation, and second progress review	Progress Presentation 2 – Aug 17, 18, 19 (2026)
Phase 7	Final system integration, documentation, and research validation	Semester 2 (Sep – Oct 2026)
Phase 8	Final project evaluation and presentation	Final Presentation – Oct 12, 13, 14 (2026)

Table 3 : Timeline Alignment

9. GANTT CHART

The table below shows the semester-wise project timeline with key milestones.
S1 = Semester 1, S2 = Semester 2. Each column represents approximately three weeks.

Task / Milestone	**S1: W1-3**	**S1: W4-6**	**S1: W7-9**	**S1: W10-12**	**S2: W1-3**	**S2: W4-6**	**S2: W7-9**	**S2: W10-12**
Literature Review and Gap Analysis	●●●●	●						
Requirements and Architecture		●●●●	●●					
NLP Conversational Travel Analyzer Development		●●●	●●●●	●●●●				
Genetic Algorithm Package Generator				●●●●	●●●●			
Real-Time Adaptive Engine Development					●●●●	●●		
External API Integration (Weather, Traffic, Transport)						●●●●	●●	
Voice-based Alert and Notification System							●●●●	●●
Dashboard Development					●●	●●●●		
System Integration and Testing							●●	●●●●
Pilot Deployment and Validation								●●●●
Documentation and Final Report							●●	●●●●
▶ MILESTONE: Proposal Presentation		●						
▶ MILESTONE: Progress Presentation 1				●				
▶ MILESTONE: Progress Presentation 2							●	
▶ MILESTONE: Final Submission								●

Table 4 : Gantt Chart

10. REFERENCES LIST

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11.APPENDICES

11.1 Appendix A – Risk Analysis

Risk Id	Risk Description	Impact	Probability	Mitigation Strategy
R1	Insufficient tourism dataset for training and testing the AI models	High	Medium	Use publicly available tourism datasets and supplement with simulated travel data
R2	Difficulty in integrating multiple system modules	Medium	Medium	Use modular architecture and API-based integration
R3	Real-time data sources such as weather or traffic APIs may be unreliable	Medium	Medium	Use multiple API sources and fallback mechanisms
R4	Optimization algorithm may require high computation time	Medium	Medium	Optimize algorithm parameters and use efficient data structures
R5	User privacy concerns when handling travel data	High	Medium	Implement secure data handling and anonymization techniques

11.2 Appendix B – Preliminary UI Mockups

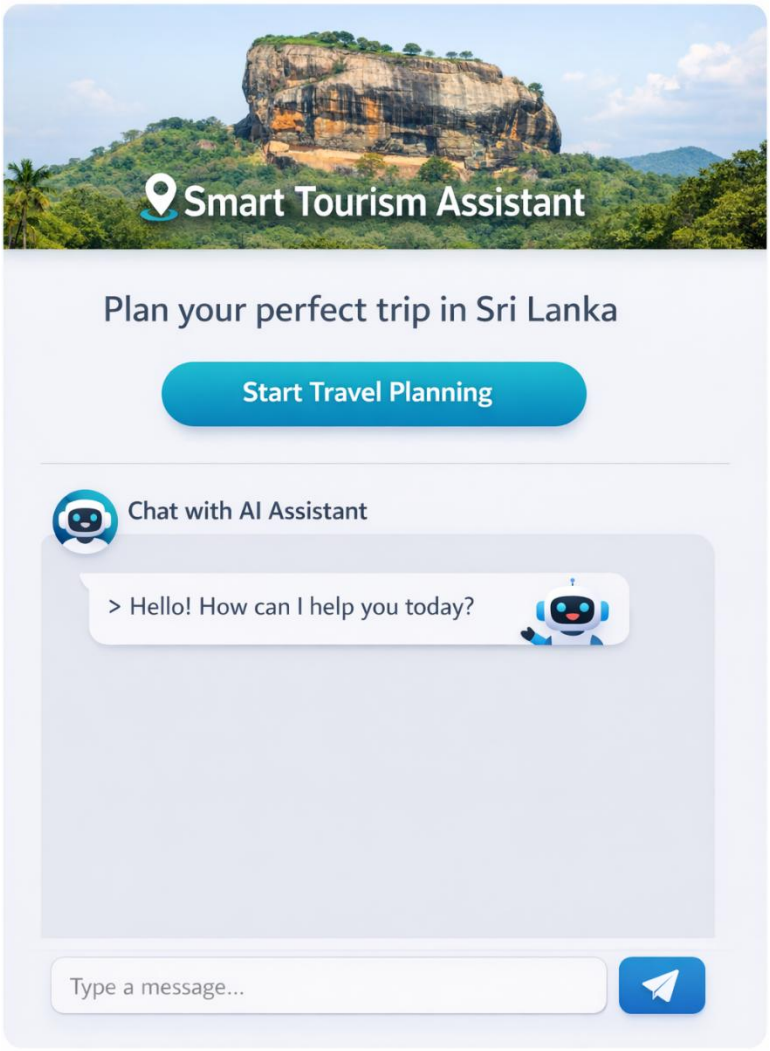


Figure 5 : Home Screen

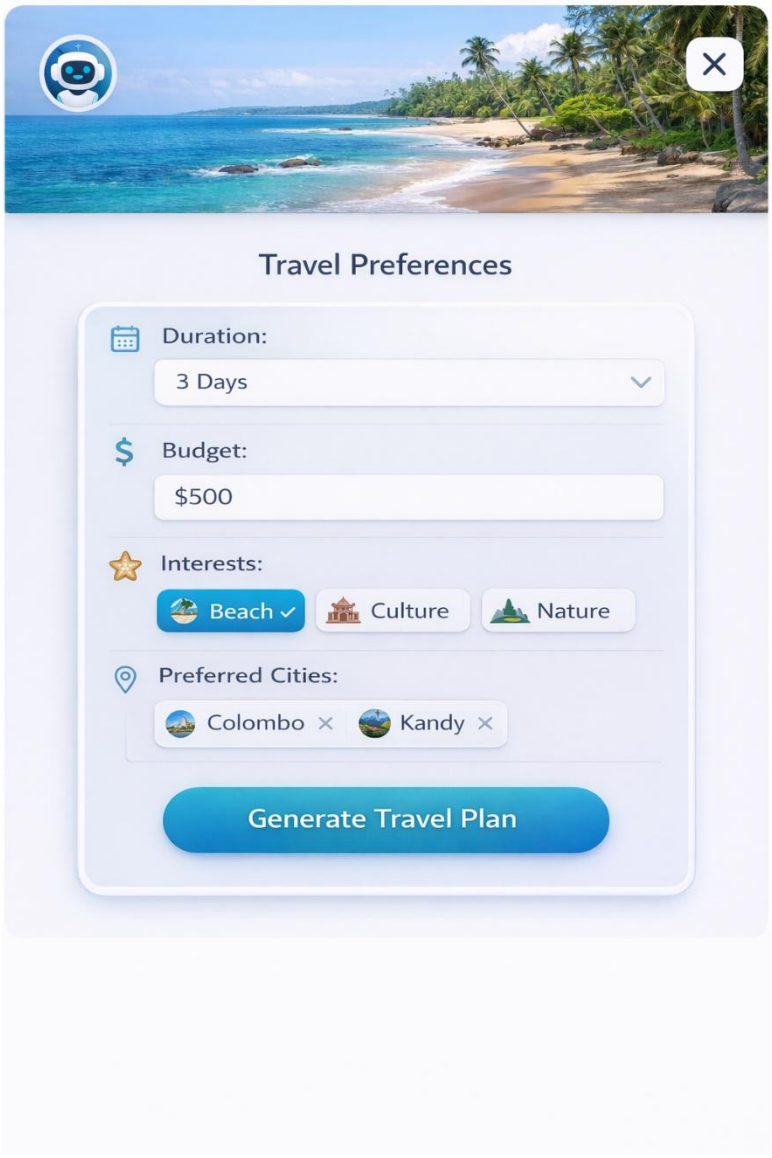


Figure 6 : Travel Preferences

11.3 Appendix C – System Workflow Diagram



Figure 7 : System Workflow Diagram

11 .4 Appendix D – Sample Travel Dataset Structure

Location	Location Name	Category	City	Latitude	Longitude	Avg Visit Time	Entry Fee	Rating
001	Sigiriya Rock Fortress	Cultural	Sigiriya	7.9570	80.7603	3 hrs	Rs.1000	4.7
002	Temple of the Tooth	Cultural	Kandy	7.2936	80.6413	2 hrs	Rs.500	4.8
003	Galle Fort	Historical	Galle	6.0329	80.2168	2 hrs	Rs.1000	4.7
004	Mirissa Beach	Beach	Mirissa	5.9483	80.4591	4 hrs	Rs.1000	4.9
005	Yala National Park	Wildlife	Yala	6.3725	81.5185	5 hrs	Rs.15 000	4.6

Table 5 : Sample Travel Dataset Structure